Detailed mathematical write-up of the Lean formalisation Lemma 1 of the Monotonicity Theorem: a constant or injective subinterval

1 Purpose and high-level structure

This document explains, in ordinary mathematical language, a Lean formalisation of *Lemma 1* from the proof of the Monotonicity Theorem for definable functions on an o-minimal structure. In the source notes (`monotonicity.tex`), the statement is recorded as a lemma occurring inside the proof of the Monotonicity Theorem, and is referred to in the running text simply as Lemma 1.

Informally, the argument splits into two cases according to whether some value of f is attained infinitely often on the interval I. If so, that infinite fibre already contains a subinterval on which f is constant. Otherwise every fibre of f on I is finite, which forces the image $f(I)$ to be infinite; choosing, for each y in an interval inside $f(I)$, the *least* preimage $g(y) \in I$, produces a one-sided inverse of f that is injective by construction. The image of g must again contain a subinterval of I, and on this last subinterval f is necessarily injective.

The Lean file discussed here (`kovacs_benedek_noel.lean`) formalises exactly this two-case argument. Unlike the companion development described in *Detailed mathematical write-up of the Lean formalisation: O-minimal structures, definable maps, images, and local properties*, the present file does not build or import a full `OMinimalStructure` object indexed by all arities n. Instead, it isolates the two non-trivial mathematical facts that the argument for Lemma 1 actually needs and records them as two explicit Lean axioms:

- `o_minimal_contains_interval`: every infinite subset of R contains an infinite subinterval of R. This is precisely the one-dimensional content of o-minimality — Definition 5.1(xi) in the companion document, together with density of the order — specialised to the case at hand.
- `infinite_image_of_injOn`: the image of an infinite set under a function that is injective on that set is again infinite. This is an ordinary fact of cardinal arithmetic; it carries no o-minimality content whatsoever.

Sections 2–6 introduce the order-theoretic vocabulary and these two axioms, together with short proofs explaining *why* each axiom is true of an actual o-minimal structure. Section 7 then gives the proof of Lemma 1 itself, following the Lean tactic proof step by step.

The main result of the file is the following statement.

Lemma 1.1 (Lemma 1: a constant or injective subinterval)**.** *Let R be a nonempty linearly ordered set, let $I \subseteq R$ be an infinite interval, and let $f : R \to R$ be a definable function. Then there is a subinterval $J \subseteq I$ such that f is either constant on J or injective on J.*

This is the Lean theorem `lemma_0_5`.

2 The ambient order, intervals, and definable functions

2.1 Intervals

Throughout, R is a type equipped with a linear order $<$ (with associated $\leq$), and all sets are subsets of R. The Lean file works with the following order-theoretic notion of an interval.

Definition 2.1 (Interval)**.** *A subset $S \subseteq R$ is an* interval*, written* $\mathrm{Int}(S)$*, if it is order-convex: for all $x, y \in S$ and all $z \in R$,*

$$x \leq z \leq y \implies z \in S.$$

In Lean this is the predicate `is_interval`.

Remark 2.2. *Definition 2.1 is the usual order-theoretic notion of a convex subset and is slightly more permissive than the inductively generated class* `FiniteUnionOfPointsAndIntervals` *of the companion document (which singles out points and* open *intervals as generators). Every point, every open interval, and every closed or half-open interval satisfies Definition 2.1. This extra generality is harmless here: the proof of Lemma 1 never inspects the internal structure of an interval, only its convexity, so the weaker and more flexible Definition 2.1 is the natural notion to use in this file.*

2.2 Definable functions

Definition 2.3 (Definable function, placeholder form)**.** *In this file, a function $f : R \to R$ is* definable *if it satisfies the always-true predicate* `DefinableFunction`$(f) :\equiv \top$*.*

Remark 2.4. *In the full theory (Definition 6.1 of the companion document), definability of $f : A \to B$ is a substantive condition: it requires A and B to be definable sets and the graph of f to be a definable subset of R^2. Because the present file does not instantiate an ambient* `OMinimalStructure`*, the predicate* `DefinableFunction` *is left as the trivial placeholder of Definition 2.3. Its purpose is purely structural: it lets the statement of Lemma 1.1 keep the same shape, f ranging over "definable" functions, that it would have in the full theory, so that* `DefinableFunction` *can later be replaced by the genuine notion of Definition 6.1 without changing the statement or proof of Lemma 1.1 at all.*

3 Constancy and injectivity on a subset

Definition 3.1 (Constant and injective on a set)**.** *Let $f : R \to R$ and $J \subseteq R$. We write*

$$\mathrm{Const}(f, J) :\iff \forall x, y \in J,\ f(x) = f(y),$$

$$\mathrm{Inj}(f, J) :\iff \forall x, y \in J,\ \big(f(x) = f(y) \implies x = y\big).$$

In Lean these are `ConstantOn` and `InjectiveOn`. With this vocabulary, Lemma 1.1 asserts exactly that some subinterval $J \subseteq I$ satisfies $\mathrm{Const}(f, J) \vee \mathrm{Inj}(f, J)$.

4 The o-minimal input: infinite sets contain an interval

Definition 4.1 (Infinite-set axiom)**.** *The file postulates, as a Lean* `axiom`*, that for every infinite $S \subseteq R$ there is a subinterval $J \subseteq S$ that is itself infinite:*

$$S \subseteq R,\ S \text{ infinite} \implies \exists J \subseteq S,\ \mathrm{Int}(J) \wedge J \text{ infinite}.$$

This is `o_minimal_contains_interval`. Although it is introduced as an axiom in this self-contained file, it is not an independent assumption about R: it is a theorem of any actual o-minimal structure, and the next lemma records its proof.

Lemma 4.2. *Let $M = (R, <, (\mathcal{S}_n)_n)$ be an o-minimal structure on a dense linear order without endpoints R (Definitions 3.1 and 5.1 of the companion document). If $S \in \mathcal{S}_1$ is infinite, then S contains an infinite open interval.*

Proof. By the one-dimensional o-minimality axiom (Definition 5.1(xi)), S is a finite union of points and open intervals,

$$S = \{a_1, \ldots, a_k\} \ \cup\ (b_1, c_1) \ \cup\ \cdots\ \cup\ (b_m, c_m).$$

The set of points $\{a_1, \ldots, a_k\}$ is finite. If there were no interval pieces ($m = 0$), S itself would be finite, contradicting the hypothesis. Hence $m \geq 1$, and S contains some open interval (b_j, c_j) with $b_j < c_j$. By density of the order (Definition 3.1(iv)), between any two elements of (b_j, c_j) there is a further element of (b_j, c_j); iterating this produces infinitely many distinct elements, so (b_j, c_j) is infinite. Since an open interval is convex, $J := (b_j, c_j)$ satisfies $\mathrm{Int}(J)$, $J \subseteq S$, and J is infinite, as required. $\square$

Remark 4.3. *Lemma 4.2 is exactly Definition 4.1, i.e. `o_minimal_contains_interval`, specialised at $n = 1$ and restated without reference to the family $(\mathcal{S}_n)_n$. Recording it as an axiom in the present file is a deliberate simplification: it avoids re-deriving, inside this file, the inductive structure of points and open intervals underlying `FiniteUnionOfPointsAndIntervals`, together with the density argument, while remaining faithful to the underlying mathematics, since the displayed implication is a genuine theorem of every o-minimal structure.*

5 A cardinality fact: injective images of infinite sets

Definition 5.1 (Injective-image axiom)**.** *The file also postulates, as a Lean `axiom`, that for $s \subseteq R$ and $f : R \to R$,*

$$s \text{ infinite},\ f \text{ injective on } s \implies f(s) \text{ infinite}.$$

This is `infinite_image_of_injOn`. In contrast with Definition 4.1, this fact carries no o-minimality content: it is a statement of ordinary cardinal arithmetic, true for an arbitrary function between arbitrary sets.

Lemma 5.2. *Let s be an infinite set and let f be injective on s. Then $f(s)$ is infinite.*

Proof. We argue by contraposition. Suppose $f(s)$ is finite. The restriction of f to s, viewed as a map $s \to f(s)$, is injective by hypothesis and surjective onto $f(s)$ by definition of the image, hence a bijection of s onto $f(s)$. An infinite set cannot be put in bijection with a finite one, so s would have to be finite as well. This contradicts the hypothesis that s is infinite. Hence $f(s)$ is infinite. $\square$

Remark 5.3. *Because Lemma 5.2 does not depend on o-minimality, it could equally well be proved inside the Lean file itself rather than postulated as an axiom (for instance by formalising the contrapositive argument above). It is recorded here as an axiom purely for economy of presentation; doing so does not weaken the formalisation of Lemma 1.1, since Lemma 5.2 is unconditionally true.*

6 A choice function for least elements

The informal proof of Lemma 1 defines an "inverse" g by $g(y) := \min\{x \in I : f(x) = y\}$. To formalise this without first proving that the relevant set always has a minimum, the Lean file uses Hilbert's choice operator.

Definition 6.1 (Least element, by choice)**.** *For $S \subseteq R$, define*

$$\mathrm{lmin}(S) := \varepsilon x.\ \big(x \in S \wedge \forall y \in S,\ x \leq y\big),$$

where ε is Hilbert's choice operator: $\varepsilon x. P(x)$ denotes an arbitrarily chosen element satisfying P if one exists, and an arbitrary element of R otherwise.

This is the noncomputable Lean definition `myInf`, built from `Classical.epsilon`. Nothing is asserted yet about $\mathrm{lmin}(S)$ in general; the next lemma isolates exactly the circumstances under which $\mathrm{lmin}(S)$ behaves as a genuine minimum.

Lemma 6.2 (Specification of lmin)**.** *If S is finite and nonempty, then $\mathrm{lmin}(S) \in S$ and $\mathrm{lmin}(S) \leq z$ for every $z \in S$; that is, $\mathrm{lmin}(S)$ is the minimum element of S.*

Proof. Since R is linearly ordered and S is finite and nonempty, S has a minimum element x_0: view S as a finite set (a `Finset`) and let x_0 be its minimum, which exists by finiteness and is characterised by $x_0 \in S$ and $x_0 \leq z$ for all $z \in S$. Thus the predicate $P(x) :\equiv x \in S \wedge \forall y \in S,\ x \leq y$ is satisfied by x_0, so $\exists x,\ P(x)$ holds. By the defining specification of Hilbert's choice operator, whenever $\exists x,\ P(x)$ holds, the chosen element $\varepsilon x. P(x)$ itself satisfies P. Hence $P(\mathrm{lmin}(S))$ holds, which is the statement of the lemma. □

This is the Lean lemma `myInf_spec`. In Lean the minimum x_0 is produced by `Set.Finite.toFinset` together with `Finset.min'`. The final step, `Classical.epsilon_spec`, then transfers this minimality to lmin.

7 Proof of Lemma 1

We now prove Lemma 1.1, following the case split used in the Lean proof of `lemma_0_5`. Fix R, I, f as in the statement, with I an infinite interval.

7.1 Splitting on the fibres of f over I

The proof distinguishes two cases, according to whether

$$\exists y \in R,\ f^{-1}(y) \cap I \text{ is infinite}$$

holds or fails. In Lean this is the case split `by_cases h_inf_preimage`.

7.2 Case 1: some fibre is infinite

Suppose $f^{-1}(y) \cap I$ is infinite for some $y \in R$. By Definition 4.1/`o_minimal_contains_interval` applied to the infinite set $f^{-1}(y) \cap I$, there is a subinterval

$$J \subseteq f^{-1}(y) \cap I$$

with $\mathrm{Int}(J)$ (the auxiliary infiniteness of J furnished by the axiom is not needed here and is discarded; in Lean this is `h_contains_interval`). Since $J \subseteq I$, J is a candidate subinterval of I. We show $\mathrm{Const}(f, J)$. For $x, z \in J$ we have $x, z \in f^{-1}(y)$, i.e. $f(x) = y = f(z)$; hence $f(x) = f(z)$. As $x, z \in J$ were arbitrary, $\mathrm{Const}(f, J)$ holds, and the disjunction $\mathrm{Const}(f, J) \vee \mathrm{Inj}(f, J)$ in Lemma 1.1 is witnessed by its left disjunct.

7.3 Case 2: every fibre is finite

Now suppose every fibre of f on I is finite:

$$\forall y \in R,\ f^{-1}(y) \cap I \text{ is finite.}$$

(In Lean, this is obtained from the negation of the Case 1 hypothesis by the tactic `push Not`, using the fact that, by definition, a set is infinite exactly when it is not finite, so the negated hypothesis simplifies directly to finiteness of every fibre; this hypothesis is recorded as `h_inf_preimage` for the remainder of the proof.)

7.3.1 The image $f(I)$ is infinite

Lemma 7.1. *Under the standing hypotheses of Case 2, $f(I)$ is infinite.*

Proof. Suppose, for contradiction, that $f(I)$ is finite. Every $x \in I$ lies in the fibre $f^{-1}(f(x)) \cap I$, so

$$I \subseteq \bigcup_{y \in f(I)} \big(f^{-1}(y) \cap I\big).$$

The index set $f(I)$ is finite by assumption, and each set $f^{-1}(y) \cap I$ in the union is finite by the standing hypothesis of Case 2. A finite union of finite sets is finite, so the right-hand side is finite, and hence I, being a subset of a finite set, is finite. This contradicts the hypothesis that I is infinite. □

This is `h_fI_inf`; in Lean the contradiction is reached directly, by assuming a proof `h_fin_img` of finiteness of $f(I)$, exhibiting I as a subset of the finite union `Set.Finite.biUnion h_fin_img (fun y _ => h_inf_preimage y)` via `Set.mem_biUnion`, and finally applying the hypothesis `hI_inf` (that I is infinite, i.e. that I is *not* finite) to the resulting proof that I is finite.

By Definition 4.1/`o_minimal_contains_interval` applied to the infinite set $f(I)$, we obtain a subinterval

$$J_{\mathrm{img}} \subseteq f(I), \qquad \mathrm{Int}(J_{\mathrm{img}}), \qquad J_{\mathrm{img}} \text{ infinite.}$$

This is `h_fI_contains_interval`; unlike in Case 1, the infiniteness of J_{img} furnished by the axiom is retained here, since it will be needed below.

7.3.2 A one-sided inverse g on J_{img}

Define

$$g : R \to R, \qquad g(y) := \mathrm{lmin}\big(\{x \in I : f(x) = y\}\big).$$

This is the Lean local definition `let g := fun y => myInf {x ∈ I | f x = y}`.

Lemma 7.2. *For every $y \in J_{\mathrm{img}}$,*

$$g(y) \in I \qquad \textit{and} \qquad f(g(y)) = y.$$

Proof. Fix $y \in J_{\text{img}} \subseteq f(I)$, so there is $x \in I$ with $f(x) = y$; thus the set $S_y := \{x \in I : f(x) = y\}$ is nonempty. By the standing hypothesis of Case 2, S_y is a subset of the finite fibre $f^{-1}(y) \cap I$ (indeed $S_y = f^{-1}(y) \cap I$), so S_y is finite. By Lemma 6.2, $\text{lmin}(S_y) = g(y)$ satisfies $g(y) \in S_y$, i.e. $g(y) \in I$ and $f(g(y)) = y$. □

This is `h_g_in_I` and `h_fg_eq`; the finiteness of each S_y is recorded once, for all y simultaneously, as `h_finite_preimage`.

7.3.3 g is injective, and its image contains an interval

Lemma 7.3. *g is injective on J_{img}.*

Proof. Let $y_1, y_2 \in J_{\text{img}}$ with $g(y_1) = g(y_2)$. By Lemma 7.2,

$$y_1 = f(g(y_1)) = f(g(y_2)) = y_2.$$

□

This is `h_g_inj`. Combining Lemma 7.3 with the infiniteness of J_{img} established above and Lemma 5.2, i.e. `infinite_image_of_injOn`, the image $g(J_{\text{img}})$ is infinite; this is `h_gJ_inf`. Applying Definition 4.1/`o_minimal_contains_interval` once more, this time to the infinite set $g(J_{\text{img}})$, yields a subinterval

$$J \subseteq g(J_{\text{img}}), \qquad \text{Int}(J)$$

(again discarding the auxiliary infiniteness of J, which is not needed further). This is `h_gJ_contains_interval`.

By Lemma 7.2, every element of $g(J_{\text{img}})$ lies in I, so $J \subseteq g(J_{\text{img}}) \subseteq I$. This inclusion is `hJ_sub_I`, and it makes J a legitimate candidate subinterval of I for Lemma 1.1.

7.3.4 Conclusion: f is injective on J

It remains to show $\text{Inj}(f, J)$. Let $x_1, x_2 \in J$ with $f(x_1) = f(x_2)$. Since $J \subseteq g(J_{\text{img}})$, there are $y_1, y_2 \in J_{\text{img}}$ with

$$g(y_1) = x_1, \qquad g(y_2) = x_2.$$

Substituting, the hypothesis $f(x_1) = f(x_2)$ becomes

$$f(g(y_1)) = f(g(y_2)),$$

and the desired conclusion $x_1 = x_2$ becomes, after the same substitution, the equivalent goal

$$g(y_1) = g(y_2).$$

By Lemma 7.2, $f(g(y_1)) = y_1$ and $f(g(y_2)) = y_2$; combined with $f(g(y_1)) = f(g(y_2))$ from above, this gives $y_1 = y_2$. Substituting back, $g(y_1) = g(y_2)$, which is exactly the goal. Hence $\text{Inj}(f, J)$ holds, and the disjunction $\text{Const}(f, J) \vee \text{Inj}(f, J)$ is witnessed by its right disjunct. This completes Case 2, and with it the proof of Lemma 1.1.

In Lean, the two substitutions described above are performed by rewriting along `hy1_eq : g y1 = x1` and `hy2_eq : g y2 = x2` (in the hypothesis `h_eq` and in the goal), the equality $y_1 = y_2$ is recorded as `h_y_eq`, obtained by rewriting along `h_fg_eq y1 hy1_in` and `h_fg_eq y2 hy2_in`, and the proof finishes with `rw [h_y_eq]`.

7.4 A remark on the hypotheses $\mathrm{Int}(I)$ and definability of f

Remark 7.4. *The Lean signature of* `lemma_0_5` *records two hypotheses, named* `_hI : is_interval I` *and* `_hf_def : DefinableFunction f`*, whose leading underscore is the Lean convention for "present for the shape of the statement, not used by name in the proof." Indeed, neither the convexity of I nor the (placeholder) definability of f is used anywhere in the argument above: only the infiniteness of I, recorded as* `hI_inf`*, is needed. This is not an oversight; it mirrors the informal proof, which likewise never uses convexity of I beyond the fact that a subset of an interval that is itself convex is again an interval — a fact that is used only to justify calling the sets J produced along the way "subintervals," not to derive any further property of f. The hypotheses are kept in the statement of Lemma 1.1 so that its signature matches the way I and f are introduced in the surrounding proof of the Monotonicity Theorem, where I is always an interval and f is always definable.*

8 Dictionary between the Lean file and this document

Lean name	Mathematical meaning
`is_interval`	The convexity predicate $\mathrm{Int}(S)$ of Definition 2.1.
`DefinableFunction`	Placeholder predicate, identically true (Definition 2.3).
`ConstantOn`	The predicate $\mathrm{Const}(f, J)$.
`InjectiveOn`	The predicate $\mathrm{Inj}(f, J)$.
`o_minimal_contains_interval`	Axiom: every infinite subset of R contains an infinite subinterval (Definition 4.1; proved from o-minimality in Lemma 4.2).
`infinite_image_of_injOn`	Axiom: the injective image of an infinite set is infinite (Lemma 5.2).
`myInf`	The choice function $\mathrm{lmin}(S) = \varepsilon x. (x \in S \wedge \forall y \in S, x \leq y)$.
`myInf_spec`	Specification of lmin on finite nonempty sets (Lemma 6.2).
`lemma_0_5`	Lemma 1 itself (Lemma 1.1).
`h_inf_preimage`	The case-split hypothesis on fibres of f over I (Section 7.1).
`h_contains_interval`	The subinterval of an infinite fibre, in Case 1.
`h_fI_inf`	Infiniteness of $f(I)$ in Case 2 (Lemma 7.1).
`h_fI_contains_interval`	The subinterval $J_{\mathrm{img}} \subseteq f(I)$, with its infiniteness.
`g`	The one-sided inverse $g(y) = \mathrm{lmin}\{x \in I : f(x) = y\}$.
`h_finite_preimage`	Finiteness of $\{x \in I : f(x) = y\}$ for every y, from the Case 2 hypothesis.
`h_g_in_I`, `h_fg_eq`	The two parts of Lemma 7.2: $g(y) \in I$ and $f(g(y)) = y$.

Lean name	Mathematical meaning
`h_g_inj`	Injectivity of g on J_{img} (Lemma 7.3).
`h_gJ_inf`	Infiniteness of $g(J_{\mathrm{img}})$.
`h_gJ_contains_interval`	The final subinterval $J \subseteq g(J_{\mathrm{img}})$.
`hJ_sub_I`	The inclusion $J \subseteq I$.
`h_y_eq`	The equality $y_1 = y_2$ used to conclude $g(y_1) = g(y_2)$ at the end of Case 2.

9 Summary in one line

The mathematical content of the Lean file is exactly the case split

$$\big(\exists y,\ f^{-1}(y) \cap I \text{ infinite}\big) \ \vee\ \big(\forall y,\ f^{-1}(y) \cap I \text{ finite}\big),$$

together with the one-dimensional o-minimality fact that every infinite subset of R contains an infinite subinterval, applied first to a single infinite fibre, and, in the remaining case, to the infinite image $f(I)$ and again to the infinite image of the resulting least-preimage function g; this assembles, in either branch, a subinterval $J \subseteq I$ on which f is constant or injective.